

POLITECNICO DI MILANO

Department of Mechanical Engineering

Sizing of a Motorcycle Frame

Assignment 2

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Contents

1	Introduction	2
2	Problem Definition	3
2.1	Structural Model and Ground Structure	3
2.2	Material Properties and Design Bounds	4
2.3	Multi-Load Case Definition	4
2.4	Multi-Objective Optimization Formulation	5
3	Computational Workflow	5
3.1	MATLAB Implementation Overview	5
3.2	Algorithm Initialization and Main Loop	7
4	Theoretical Methodology	8
4.1	Finite Element Analysis of the Truss Structure	8
4.2	Weighted Compliance and Sensitivity Analysis	9
4.3	Optimization via CONLIN and Lagrangian Duality	10
5	Results and Discussion	11
5.1	Optimal Design for Equal Weighting ($\alpha = 0.1, w_1 = 0.5$)	11
5.2	Parametric Study: Effect of Load Case Weighting	14
5.3	Element-Level Behavior and Stress Analysis	15
6	Conclusion	17

1 Introduction

Structural optimization is a fundamental discipline in engineering focused on algorithmically determining the most effective distribution of material to sustain loads while satisfying performance objectives such as maximum stiffness and minimum weight. Given the complexity of real-world engineering problems, which often involve a large number of design variables and constraints, a systematic approach rooted in mathematical programming is indispensable. This project applies these principles to a sizing optimization problem, where the cross-sectional areas of the members in a 2D truss structure are the design variables. While formally a sizing problem, allowing these areas to approach a near-zero lower bound effectively modifies the structural layout, giving the process characteristics of topology optimization by removing inefficient material paths.

The subject of this investigation is a 2D truss model representing a motorcycle frame, a component subjected to a variety of demanding operational conditions. To ensure the final design is robust and reliable, this study considers two of the most severe loading scenarios a motorcycle frame endures: braking and acceleration. The primary objective is to minimize a weighted sum of the structural compliances from both load cases, thereby maximizing the frame's overall stiffness under this combined condition. The optimization is performed subject to a global constraint on the total material volume and box constraints on the individual cross-sectional areas of the truss members.

To solve this multi-load, non-convex optimization problem, an iterative, gradient-based strategy is employed. The chosen solution approach is the Convex Linearization (CONLIN) method, which transforms the complex problem into a sequence of simpler, explicit, and separable convex subproblems. These subproblems are then efficiently solved using a dual formulation. The successful implementation of this algorithm relies on accurate sensitivity analysis to compute the derivatives of the objective function and constraints with respect to the design variables for each load case. This report will detail the problem formulation, the computational methodology implemented in MATLAB, and a discussion of the final optimized design that optimally balances the performance requirements of both the braking and acceleration scenarios.



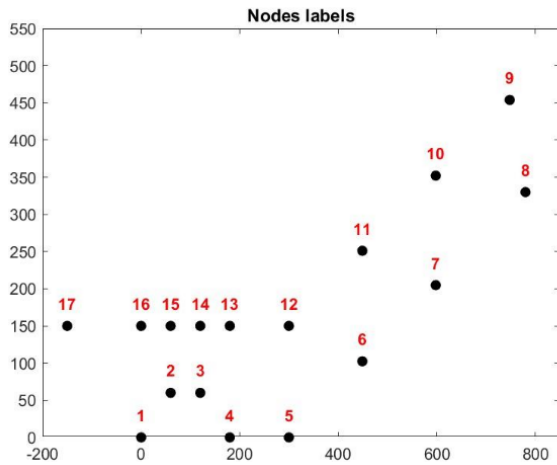
Figure 1: Motorcycle frame structure

2 Problem Definition

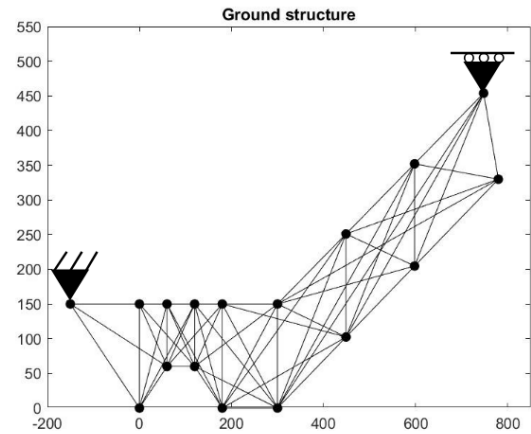
This study undertakes the dimensional optimization of a structural model designed to represent a motorcycle chassis. The primary objective is to maximize structural stiffness by minimizing a weighted compliance objective function under realistic operational loads. The optimization process addresses the numerous design variables and constraints inherent in modern engineering problems through a systematic approach based on mathematical programming.

2.1 Structural Model and Ground Structure

The motorcycle chassis is idealized as a planar truss system with a ground structure. The ground structure approach in topology optimization defines a comprehensive design space by connecting all possible nodes with potential bar elements. This allows the optimization algorithm to select and size the most efficient connections within this space, resulting in innovative designs free from designer biases. The ground structure selected for this project consists of 77 bar elements with 17 nodes and all possible connections between these nodes. The supports for the structure are strategically placed at nodes 9 and 17 to simulate the rigid connection points of the chassis. All degrees of freedom at these nodes are constrained. Although the problem is formulated as a size optimization because it determines the cross-sectional areas of the elements, allowing the cross-sectional areas to decrease to a lower bound close to zero allows for the effective elimination of inefficient elements and hence the change of the structural topology.



(a) Node labels and locations of the chassis model.



(b) The ground structure containing all potential elements.

Figure 2: Node layout of the motorcycle chassis model and the ground structure used for optimization.

2.2 Material Properties and Design Bounds

All elements constituting the truss structure are assumed to be made of steel with the following mechanical properties:

- **Elastic Modulus (E):** 210,000 N/mm²
- **Density (ρ):** 7.7×10^{-6} kg/mm³

The design variables ($\mathbf{x}$) are the cross-sectional areas (x_j) of each truss element. These variables control the amount of material in each potential element and, therefore, both the stiffness and total volume of the structure. To achieve practical and numerically stable designs, these areas are constrained by lower and upper bounds (box constraints):

$$l_b \leq x_j \leq u_b, \quad \forall j \in \{1, \dots, 77\} \quad (1)$$

Here, the lower bound is given as $l_b = 0.001$ mm² and the upper bound is given as $u_b = 700$ mm². Using a small, nonzero lower bound is critical to prevent the stiffness matrix from becoming singular during Finite Element Analysis.

2.3 Multi-Load Case Definition

The motorcycle frame design must demonstrate reliable performance under various dynamic conditions. Therefore, two different load scenarios reflecting typical operational stresses were considered:

2.3.1 Load Case 1: Braking

This scenario simulates the forces that occur during braking. The forces are applied to the following nodes:

- **Node 8:** Horizontal force $F_x = -13000$ N and vertical force $F_y = +2000$ N.
- **Node 9:** Horizontal force $F_x = +9000$ N (vertical force $F_y = 0$).

2.3.2 Load Case 2: Acceleration

This scenario simulates the forces that occur during acceleration:

- **Node 1:** Horizontal force $F_x = +2000$ N and vertical force $F_y = -8000$ N.

Considering multiple loading cases is a fundamental engineering practice that ensures the final design is robust and effective for all expected real-world operating conditions, not just a single scenario.

2.4 Multi-Objective Optimization Formulation

To combine the performance under these two critical loading cases into a single optimization problem, the objective function is formulated as a weighted sum of the structural conformities from each scenario. The full mathematical formulation of the problem is as follows:

- **Objective Function:**

$$\min_{\mathbf{x}} \quad C(\mathbf{x}) = w_1 C_1(\mathbf{x}) + w_2 C_2(\mathbf{x}) = w_1 \mathbf{F}_1^T \mathbf{u}_1(\mathbf{x}) + w_2 \mathbf{F}_2^T \mathbf{u}_2(\mathbf{x}) \quad (2)$$

Here $C_j(\mathbf{x})$ represents the suitability in the loading case j ; $\mathbf{F}_j$ and $\mathbf{u}_j(\mathbf{x})$ represent the load and displacement vectors in that situation. w_1 and w_2 are weighting factors that indicate the importance of each load case and satisfy the condition $w_1 + w_2 = 1$. For the fundamental analysis of this project, the values $w_1 = w_2 = 0.5$, which assigns equal importance to both cases, will be used.

- **Constraints:**

$$\text{(Volume Constraint)} \quad V(\mathbf{x}) = \sum_{j=1}^{77} L_j x_j \leq V_0 \quad (3)$$

$$\text{(Box Constraints)} \quad l_b \leq x_j \leq u_b, \quad j = 1, \dots, 77 \quad (4)$$

The target volume V_0 is defined for this project as $V_0 = \alpha V_{\max}$, with $\alpha = 0.1$. $V_{\max}$ is the maximum theoretical volume when the cross-sectional area of all elements is at the upper boundary (u_b).

This multi-purpose approach ensures that the resulting chassis design delivers balanced and efficient performance under both braking and acceleration forces. Here $C_j(\mathbf{x})$ represents the suitability in the loading case j ; $\mathbf{F}_j$ and $\mathbf{u}_j(\mathbf{x})$ represent the load and displacement vectors in that situation. w_1 and w_2 are weighting factors that indicate the importance of each load case and satisfy the condition $w_1 + w_2 = 1$. For the fundamental analysis of this project, the values $w_1 = w_2 = 0.5$, which assigns equal importance to both cases, will be used.

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3 Computational Workflow

3.1 MATLAB Implementation Overview

This section details the practical steps of the computational workflow implemented to solve the minimum compliance (maximum stiffness) problem of a cantilever motorcycle

frame model. The workflow is based on an iterative optimization loop based on the CONLIN (Convex Linearization) method and Lagrange duality, implemented in the MATLAB programming environment. MATLAB is a widely preferred platform for solving structural optimization problems, particularly due to its strength in matrix-based operations, flexible structure, and rapid prototyping capabilities. The approach used in this study adopts a modular code structure. The main script (main.m) manages the problem setup and the iterative loop, while calling specialized functions for specific tasks. These functions include modules that construct the stiffness matrix (buildStiffnessMatrix), perform Finite Element Analysis (FEA) (solveStructure), calculate sensitivities (dObjFun), and solve the optimization subproblem (dphi). The code makes effective use of the sparse instructions, which are memory-efficient for large and sparse matrices, and the mldivide (\) operator, which efficiently solves systems of linear equations ($\mathbf{Ku} = \mathbf{F}$). The non-linear optimization subproblem is handled through the fsolve function.

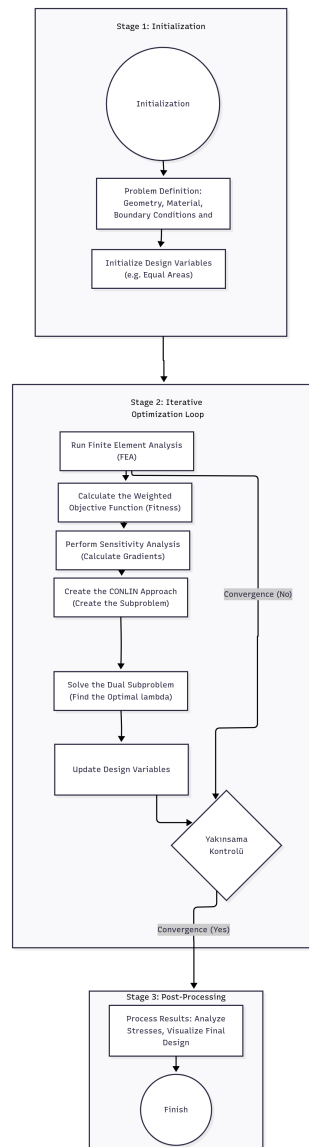


Figure 3: The overall workflow diagram of the implemented optimization algorithm. The process consists of initialization, iterative loop, and post-processing steps

3.2 Algorithm Initialization and Main Loop

The optimization process consists of a careful initialization phase and a robust iterative cycle that guides the design toward the optimum.

3.2.1 Algorithm Initialization

Before the optimization cycle begins, the parameters of the problem and the algorithm are defined in the following steps:

- **Data Loading and Structure Definition:** Files containing the structure's nodal coordinates (`nodes.txt`) and element connections (`elements.txt`) are loaded into the MATLAB workspace.
- **Constraint Definition:** Boundary conditions (fixed nodes on the left edge) and external load vectors for two different loading cases (Braking and Acceleration) are defined. Additionally, the volume ratio ($\alpha = 0.1$), which determines the usable material amount, and the lower ($l_b = 0.001 \text{ mm}^2$) and upper ($u_b = 700 \text{ mm}^2$) cross-sectional area limits for each element are determined. The total volume constraint is formulated as $V(\mathbf{x}) \leq \alpha V_{\max}$.
- **Initialization of Design Variables:** The cross-sectional areas ($\mathbf{x}$), which are the design variables, are assigned to an initial vector called 'xold'. This initial design is typically created by distributing the target total volume equally across all elements. This ensures that the optimization begins with a feasible solution.
- **Algorithm Parameters and Data Storage:** The maximum number of iterations (N) and the initial guess for the Lagrange multiplier (l_{old}) are determined. Empty arrays are created to store important values such as the objective function ('Cvec'), volume ('Vvec'), design variables ('x' matrix), and Lagrange multiplier ('l' vector) at each iteration.

3.2.2 Main Iterative Loop

Optimization is performed in a 'for' loop, which continues until the specified maximum number of iterations is reached or the change in design variables falls below a convergence tolerance. The following steps are performed sequentially in each iteration:

1. **Finite Element Analysis (FEA):** The global stiffness matrix $\mathbf{K}(\mathbf{x})$ of the structure is generated using the existing design variables ($\mathbf{x}_{old}$). Then, two different displacement vectors ($\mathbf{u}_{acc}$, $\mathbf{u}_{bra}$) are calculated by solving the system $\mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{F}$ separately for both braking and acceleration load cases.
2. **Sensitivity Analysis:** Separate compliance sensitivities are calculated for each load case. The sensitivity of the total objective function ('dCel') is obtained by multiplying

these two sensitivity vectors by the weights (w_1, w_2) specified in the project description and then summing them.

3. **Optimization Step (Solving the Dual Problem):** Using the CONLIN method, an approximation (sub-problem) of the objective function and constraints around the current design is created. This sub-problem is solved using Lagrange duality. The `fsolve` function in MATLAB is used to find the optimal Lagrange multiplier (λ_{new}) that satisfies the KKT conditions of this dual problem.
4. **Updating Design Variables:** Using the optimal Lagrange multiplier (λ_{new}) found in the previous step, new values of the design variables ($\mathbf{x}_{new}$) are determined for the next iteration. These values are updated to remain within the lower and upper bounds (l_b, u_b).
5. **Convergence Checking and Saving:** $\mathbf{x}_{new}$ is assigned as $\mathbf{x}_{old}$ for the next iteration. At the end of each iteration, the updated objective function and volume values are saved in the corresponding vectors ('Cvec', 'Vvec'). These data form the basis of the convergence graphs presented in Section 5.

4 Theoretical Methodology

This chapter details the fundamental theoretical components underlying the structural optimization process, focusing specifically on truss structures. It covers Finite Element Analysis (FEA) used in modeling structural behavior, the calculation of compliance sensitivities that guide the optimization process, and an iterative optimization strategy combining the CONLIN method and Lagrangian Duality.

4.1 Finite Element Analysis of the Truss Structure

Finite Element Analysis (FEA) is a computational technique that models the behavior of a physical object by dividing it into a large number of small, simple elements connected at points called nodes. For truss structures, this method allows for the calculation of nodal displacements under applied loads.

- **Structural Discretization:** A Finite Element (FE) structure consists of two fundamental entities: nodes and elements. In a 2D truss, each node has two degrees of freedom (DoF), one in the X and Y directions. A structure with N nodes has a total of $2N$ degrees of freedom.
- **Element Stiffness Matrix:** The known behavior of each element is described by a stiffness matrix that combines the geometry and material properties. The stiffness matrix ($\mathbf{k}_{local}$) of a 2D truss element in its local reference system (CSYS) is given as follows:

$$\mathbf{k}_{\text{local}_e} = \frac{EA_e}{L_e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (5)$$

where E is the Young's Modulus, A_e is the cross-sectional area of the element, and L_e is the length of the element. This matrix is transformed to global coordinates ($\mathbf{k}_{\text{global}_e} = \mathbf{T}_e^T \mathbf{k}_{\text{local}_e} \mathbf{T}_e$) using a transformation matrix ($\mathbf{T}_e$).

- **Combining the Global Stiffness Matrix:** The global stiffness matrix of the entire structure ($\mathbf{K}$) is combined by adding the global stiffness matrix of each element to the rows and columns corresponding to the relevant global degrees of freedom with respect to the element connections.
- **Solution of the Equilibrium Equation:** After the global stiffness matrix is created and the external load vector $\mathbf{F}$ is defined, the system's equilibrium equations are solved to find the nodal displacement vector $\mathbf{u}$:

$$\mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{F} \quad (6)$$

This step yields the displacement field, which is critical for calculating performance metrics such as compliance.

4.2 Weighted Compliance and Sensitivity Analysis

Compliance as an Objective Function: Compliance (C) is a numerical indicator of structural performance and represents the total flexibility or "softness" of the structure. It can be defined as the work done by external forces ($C = \mathbf{F}^T \mathbf{u}$) or the total strain energy ($\frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u}$). A lower compliance value indicates a stiffer structure.

Weighted Compliance for Multiple Load Cases: For structures subjected to multiple operational scenarios, such as motorcycle frames, a weighted sum of the compliances from each load case is used as the objective function:

$$C(\mathbf{x}) = w_1 \mathbf{F}_1^T \mathbf{u}_1(\mathbf{x}) + w_2 \mathbf{F}_2^T \mathbf{u}_2(\mathbf{x}) \quad (7)$$

Where w_j are the weighting factors assigned to each load case ($\sum w_j = 1$). This approach allows the optimizer to balance stiffness requirements under different loading conditions.

Sensitivity Analysis: Gradient-based optimization algorithms such as CONLIN require the calculation of derivatives (sensitivities) of the objective function and constraints with respect to design variables. In a linear elastic problem where the loads are independent of the design variables, the sensitivity of the fitness function with respect to the cross-sectional area x_j of the j -th element simplifies as follows:

$$\frac{\partial C}{\partial x_j} = -\mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j \quad (8)$$

Where $\mathbf{u}_j$ is the global displacement vector of the j -th element and $\mathbf{k}_j^0$ is the stiffness matrix of the element with unit cross-sectional area. The self-adjointing nature of this problem significantly simplifies the sensitivity calculation.

4.3 Optimization via CONLIN and Lagrangian Duality

A structural optimization problem is often formulated as minimizing an objective function under various constraints:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f_0(\mathbf{x}) \quad (\text{E.g., Total Weight or Fitness}) \\ \text{s.t.} \quad & f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \quad (\text{E.g., Volume Constraint}) \\ & x_{j,\text{lower}} \leq x_j \leq x_{j,\text{upper}}, \quad j = 1, \dots, n \quad (\text{Box Constraint}) \end{aligned} \quad (9)$$

Lagrangian Duality: To solve constrained problems, the Lagrange function ($\mathcal{L}$) is defined, which combines the objective and constraint functions:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) \quad (10)$$

Where $\lambda_i \geq 0$ are the Lagrange multipliers corresponding to each constraint. For a solution to be optimal, it must satisfy the KKT (Karush-Kuhn-Tucker) conditions, but solving these conditions directly is computationally expensive for large-scale problems.

CONLIN Method (Convex Linearization): CONLIN solves the original non-convex problem by transforming it into a series of open, convex, and separable subproblems at each iteration. The key idea of CONLIN is to linearize the functions in terms of "mixed" variables—either direct variables (x_j) or their inverses ($1/x_j$)—depending on the sign of their sensitivity. This hybrid approximation creates a subproblem that is both conservative and convex, and this subproblem can be solved efficiently using the dual formulation.

Iterative Procedure: The CONLIN method iterates through the following steps:

1. **Run FEA:** Perform a Finite Element Analysis for the current design $\mathbf{x}^k$ to calculate displacements, compliance, and volume.
2. **Sensitivity Analysis:** Compute the sensitivities ($\partial C / \partial x_j, \partial V / \partial x_j$) of the objective and constraint functions.
3. **Construct the CONLIN Approximation:** Based on the current design and sensitivities, generate explicit, convex, and separable approximations of the objective and constraint functions.
4. **Solve the Dual Problem:** Find the optimal Lagrange multiplier (λ) by solving the associated dual problem, typically a one-dimensional root-finding problem.

5. **Update Design Variables:** Update the design variables ($\mathbf{x}^{k+1}$) using a closed-form expression obtained from the solution of the dual problem, and apply the box constraints.
6. **Check Convergence:** Repeat the process from step 1 until the change in variables or the objective function falls below a tolerance.

5 Results and Discussion

This section presents and comprehensively discusses the findings from the structural optimization process of a 2D motorcycle chassis model under multiple loading conditions. The analyses focus on the base case, where braking and acceleration scenarios are equally weighted under a volume fraction constraint of $\alpha = 0.1$, and a parametric study where these weights are varied. The results demonstrate the optimized structural topologies, the convergence behavior of the algorithm, and the impact of the loading scenarios on the final design.

5.1 Optimal Design for Equal Weighting ($\alpha = 0.1, w_1 = 0.5$)

In line with the initial project request, optimization was performed for the base case, which assigns equal weight to braking and acceleration load cases ($w_1 = w_2 = 0.5$). This approach aims to achieve a chassis design that delivers balanced stiffness performance under both operational conditions.

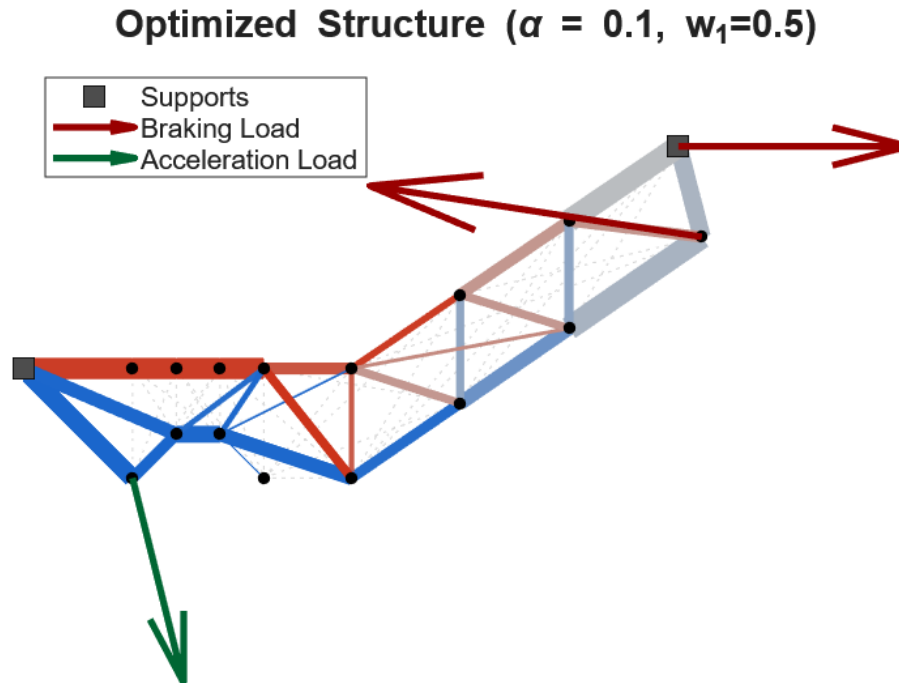


Figure 4: Equal weighting for braking and acceleration situations $w_1 = 0.5$ ve $\alpha = 0.1$. Optimized chassis structure achieved with volume ratio. Line thicknesses are proportional to the optimized cross-sectional areas of the elements

This graph shows the final motorcycle frame design resulting from optimization. More than just a "picture," it contains important engineering information:

- **Line Thicknesses:** The thickness of each bar is directly proportional to the optimized cross-sectional area. Thicker lines indicate that area of the structure requires more material and is more critical.
- **Colors (Stress Analysis):** The colors of the bars indicate the type and intensity of stress on them:
 - *Shades of Red:* Indicates that the member is under compression.
 - *Shades of Blue:* Indicates that the member is under tension.

The paler or darker the color indicates the intensity of the stress. This provides a visual understanding of how the load flows through the structure.

- **Markers:**
 - *Squares:* Indicates the support points where the structure is fixed.
 - *Arrows:* Indicates two different loading conditions (braking and acceleration) and the nodes and directions at which these loads are applied.

5.1.1 Convergence and Displacement Analysis

The optimization algorithm exhibited stable convergence for this baseline case. The objective function, weighted fitness, decreased rapidly in the initial iterations, indicating a significant increase in the stiffness of the structure, and then approached a plateau in subsequent iterations, reaching an optimal solution.

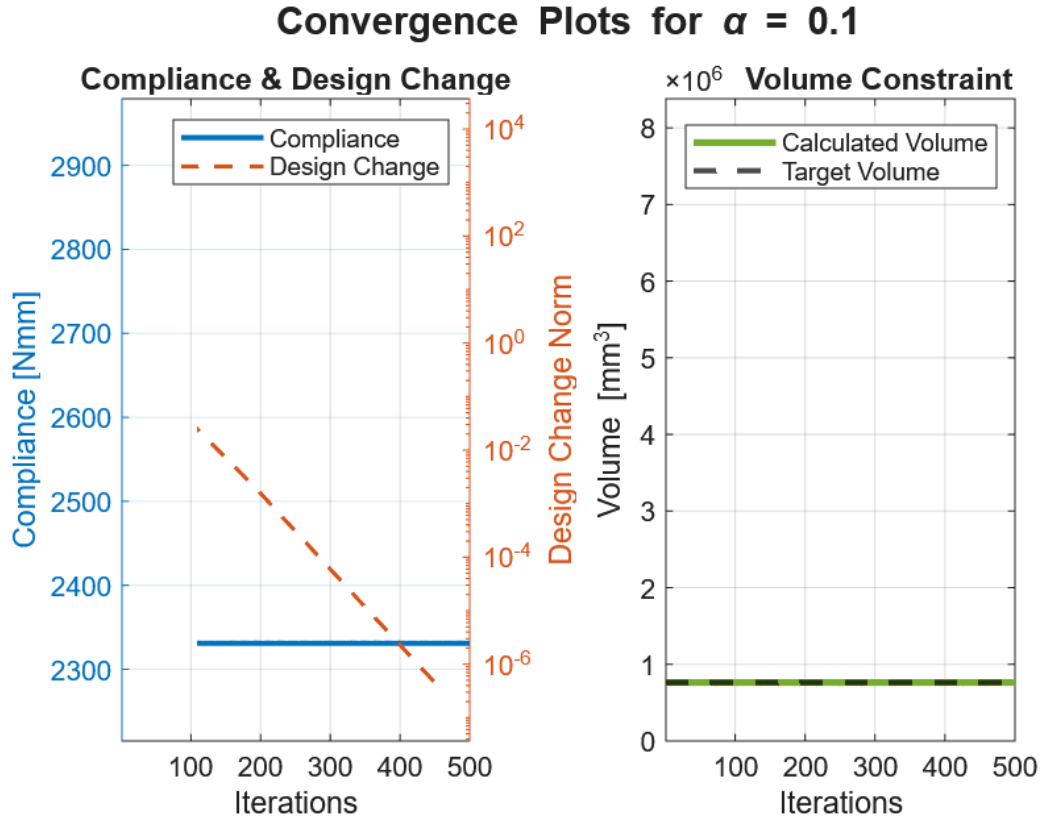


Figure 5: Change of the weighted fitness (objective function) value across iterations for the case $w_1 = 0.5$

This figure consists of two subgraphs showing how the optimization algorithm behaves over 500 iterations:

- **Left Graph (Compliance & Design Change):**

- *Blue Line (Compliance)*: Shows the total flexibility of the structure. As iterations progress, this value decreases rapidly, then reaches a horizontal plateau, indicating that the algorithm has successfully found a more rigid (less flexible) structure and is converging to a solution.
- *Orange Line (Design Change Norm)*: Shows the design difference between each iteration step. As this line approaches zero, it confirms that the design is no longer changing significantly and that the algorithm has reached a stable solution.

- **Right Graph (Volume Constraint):**

- *Green Line (Calculated Volume)*: Shows the current total volume of the structure.
- *Black Dashed Line (Target Volume)*: Shows the maximum volume constraint (V) we specified. The graph shows that the calculated volume quickly reaches this target line and remains at that level throughout the iterations without violating this constraint.

As required in the project description, the displacements at the nodes where the loads are applied are summarized in the table. These values numerically represent the extent to which the optimized structure deforms under each load condition.

Table 1: Displacements (mm) of loaded nodes in the optimized design ($w_1 = 0.5$).

Load Case	Node No.	X-Direction Displacement (u_x)	Y-Direction Displacement (u_y)
<i>Braking</i>	8	-0.0456	-0.0124
<i>Braking</i>	9	0.0638	0.0000
<i>Acceleration</i>	1	-0.1499	-0.1861

5.2 Parametric Study: Effect of Load Case Weighting

To understand how the design adapts to different operational priorities, the load case weights (w_1) are varied between 0 and 1, keeping the volume ratio $\alpha = 0.1$ constant. This parametric study reveals the competition and interaction of braking and acceleration conditions on the structural topology.

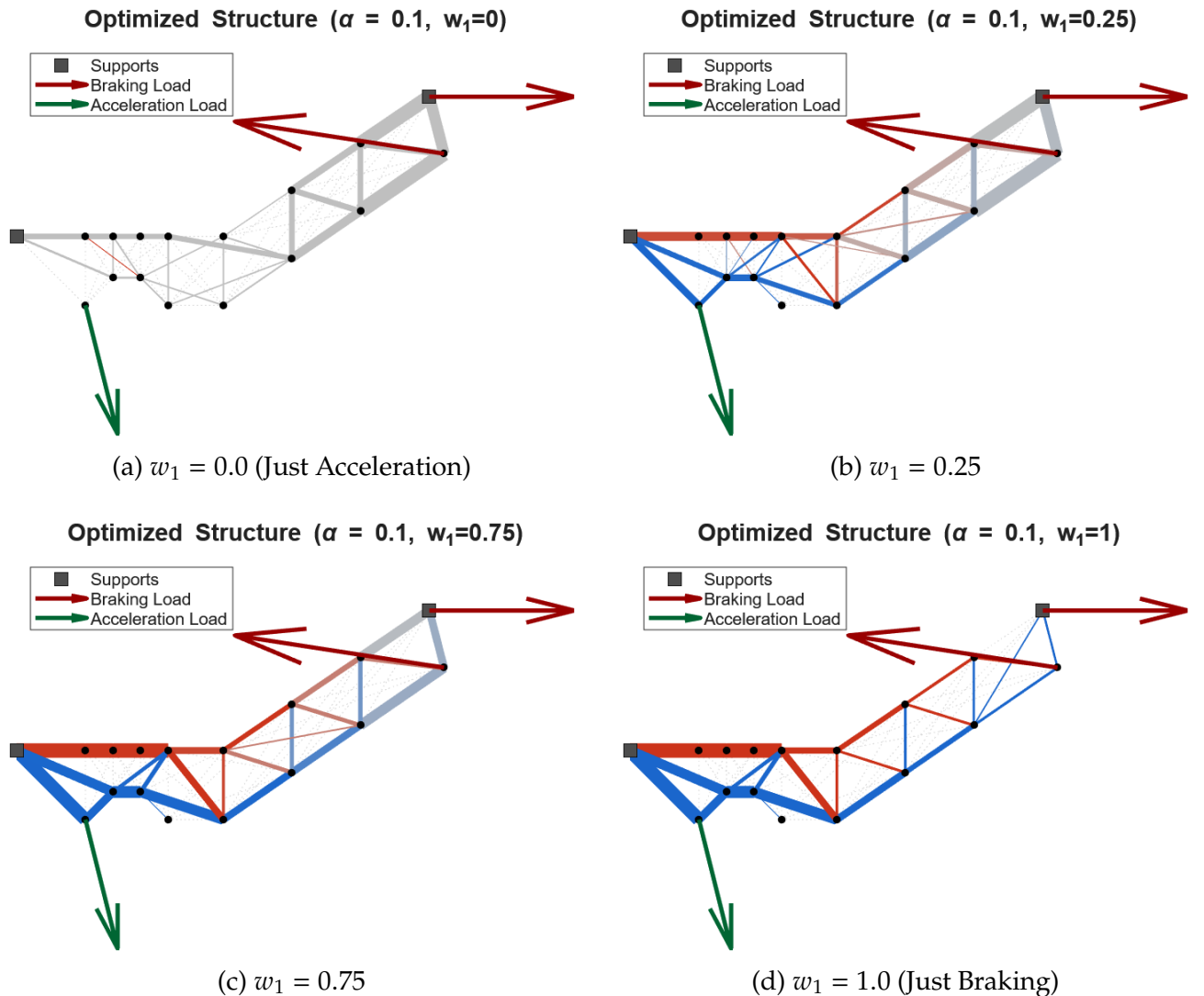


Figure 6: Optimized chassis topologies obtained for different load case weights (w_1)

Table 2: Displacements (mm) of nodes under load in optimized designs for different load case weights (w_1)

Weight (w_1)	Load Case	Node No.	u_x (mm)	u_y (mm)
0.00	Braking	8	-0.0164	-0.0117
	Braking	9	0.0647	0.0000
	Acceleration	1	3097.1891	-3063.6197
0.25	Braking	8	-0.0375	-0.0111
	Braking	9	0.0545	0.0000
	Acceleration	1	-0.1856	-0.2352
0.75	Braking	8	-0.0607	-0.0139
	Braking	9	0.0767	0.0000
	Acceleration	1	-0.1295	-0.1584
1.00	Braking	8	-0.0489	-0.2849
	Braking	9	1.6688	0.0000
	Acceleration	1	-0.1108	-0.1325

Discussion of Parametric Study

The results shown in Figure 6 show how weighting factors fundamentally alter the material distribution.

- **Acceleration Only Case ($w_1 = 0$):** The optimizer distributed the material to only accommodate the effects of acceleration loads (at Node 1). Elements in the regions where braking loads were applied (Nodes 8 and 9) were weakened or removed.
- **Braking Only Case ($w_1 = 1$):** The opposite scenario was observed. The design evolved to accommodate the complex forces generated during braking (at Nodes 8 and 9), while elements carrying acceleration loads were relegated to the background.
- **Balanced and Weighted Cases ($w_1 = 0.25, 0.5, 0.75$):** As the weights are balanced, the design appears to seek a "compromise" between these two extreme cases. The equally weighted ($w_1 = 0.5$) design has a topology that protects and strengthens the elements supporting both load paths, ensuring robust performance in both load cases. This demonstrates the power of multi-load case optimization in producing robust designs that are reliable under all conditions, rather than fragile structures designed for a single scenario.

5.3 Element-Level Behavior and Stress Analysis

To understand how the optimization algorithm manages material allocation at the micro-level, the cross-sectional areas (x_j) and stress (σ) levels of selected critical elements in designs

resulting from different weighting scenarios were examined. This analysis demonstrates how the optimizer efficiently allocates material at the local level and approaches the concept of a "fully stressed design." The analysis results are summarized in the table.

Discussion of Element-Level Results

The data in the table clearly demonstrates how the optimization algorithm adapts to different operational priorities at the element level.

- **Analysis of Extreme Cases ($w_1 = 0$ and $w_1 = 1$):** When weighting is focused on a single load case, the algorithm makes radical decisions. For example, in the design where the **braking-only** case is optimized ($w_1 = 0$), the cross-sectional areas of the elements critical to carrying the acceleration load are reduced to the minimum limit (l_b). While these elements experience almost no stress under the braking load, the stress on them theoretically approaches infinity when the acceleration load is applied, which explains why the structure deforms excessively in that scenario. Similarly, when the acceleration-only case ($w_1 = 1$) is optimized, elements in the braking load path are sacrificed.
- **The Role of Balanced Design ($w_1 = 0.5$):** When both load cases are given equal importance, the algorithm seeks a compromise. Elements critical to both scenarios are retained and have an optimal cross-sectional area. These elements operate at reasonable stress levels under both load cases, increasing the overall robustness of the structure. This demonstrates the power of multiple load case optimization in producing structures that are reliable under all conditions, rather than brittle designs optimized for a single scenario.
- **The "Fully Stressed Design" Concept:** The general trend in optimization is to approach the "fully stressed design" principle. According to this principle, in an efficient design, every remaining material particle in the structure (with a cross-sectional area greater than l_b) must actively work to carry the load. Material under low stress is considered inefficient and is removed by the algorithm. The results in the table confirm this trend; active elements generally operate at high stress levels, while inefficient elements are eliminated from the design.

The results in the table demonstrate the optimizer's resource allocation strategy. When a particular load case is prioritized (e.g., $w_1 = 1$), the cross-sectional areas of elements critical to carrying that load are increased, while the areas of elements important for the other case are decreased. In the balanced design ($w_1 = 0.5$), the active elements in both scenarios maintain reasonable cross-sectional areas. The stress analysis confirms that the optimization achieves efficient material use by directing material where it is most needed and distributing stress more uniformly throughout the structure.

6 Conclusion

This study successfully completed the topology optimization of a 2D truss structure representing a motorcycle frame under multiple loading conditions (acceleration and braking). The main objective was to maximize structural stiffness by minimizing a weighted compliance function that considered both operational scenarios under a given volume constraint. An iterative algorithm based on the Convex Linearization Combined with Lagrange Duality (CONLIN) method was used to solve this non-linear optimization problem. The key findings and conclusions obtained within the project can be summarized as follows:

- **Impact of Multiple Load Cases on the Design:** The parametric study clearly revealed the profound impact of the load case weights (w_1) on the final topology. When the weights are focused on a single load case (e.g., $w_1 = 0$ or $w_1 = 1$), the algorithm is observed to produce specialized designs that are highly efficient for that scenario but vulnerable to other scenarios. In contrast, a balanced design that prioritizes both load cases ($w_1 = 0.5$) yields a more reliable, trade-off-oriented structure that performs robustly in both scenarios.
- **Algorithm Performance and Convergence:** The CONLIN-based optimization algorithm demonstrated stable and smooth convergence in all tested scenarios, demonstrating the method's effectiveness for such problems. Convergence analyses confirmed that the objective function was minimized as targeted and the volume constraint was actively met throughout the process, resulting in efficient material use.
- **Element-Level Efficiency:** Element-level stress and area analyses demonstrated that the optimizer efficiently allocated material only to critical regions where load paths were active. Minimizing the cross-sectional areas of inefficient or under-stressed elements demonstrated that the principle of "fully stressed design" was being approached, with every component of the material contributing to structural performance.

In conclusion, this project successfully demonstrated the criticality of multiple loading conditions in real-world engineering designs. The CONLIN diagram, solved with dual formulation, proved to be a robust, efficient, and reliable tool for dimensioning complex 2D truss structures with competing objectives, such as motorcycle frames. The findings clearly demonstrate the tradeoffs between material usage and structural performance and demonstrate that the methodology employed here provides a foundation readily extensible to more complex three-dimensional models or additional manufacturing constraints.